

PART B.

Q1) Season, room type & stays are taken as input. Depending on season, appropriate and room type, and rate is selected. IF stay is greater than 7, then a discount is applied. Total hotel revenue & final rate is produced.

PAC.

Given Data  
Season room type

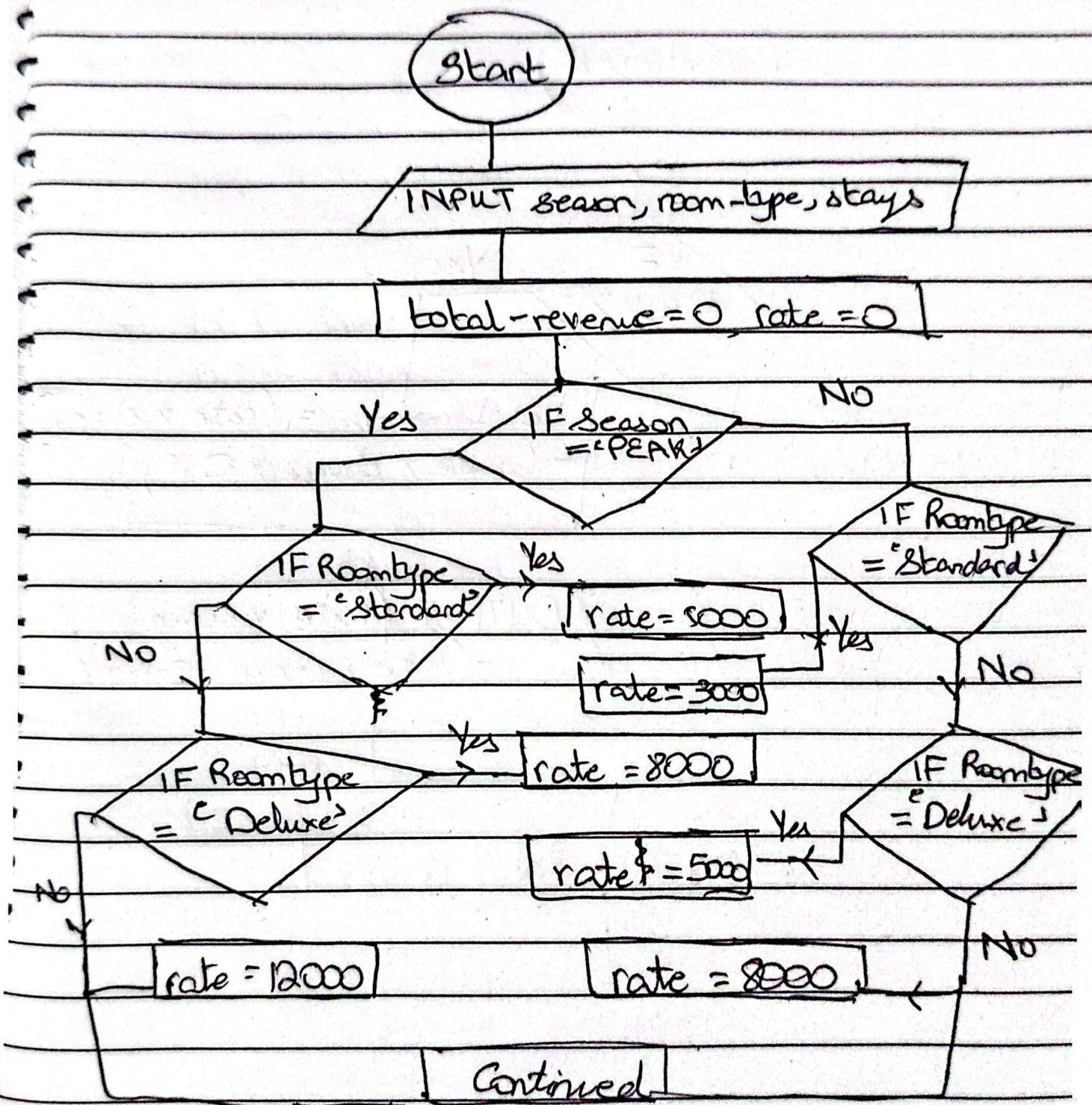
Required Results.  
final price total revenue.

Processing Required.  
Room cost, no. of stays  
final price, discount

" Solution Alternative  
Nested loops

IPO Chart

| Input            | Process.                       | Output                    |
|------------------|--------------------------------|---------------------------|
| Season Room type | Discount, Rate based on season | final price total revenue |



Continued



No

Yes

total revenue =  
(rate  $\times$  stays)

total revenue = (rate  $\times$  stays)  
- (rate  $\times$  stays  $\times$  0.15)

OUTPUT (rate  
 $\times$  stays)

OUTPUT (rate  $\times$  stays)  
- (rate  $\times$  stays  $\times$  0.15)

OUTPUT Total - Revenue

END

INPUT Season, room-type, stays.  
 total-revenue = 0, rate ← 0

IF Season == 'Peak' THEN

IF room-type = 'Standard' THEN

rate = 5000.

ELSE IF room-type = 'Deluxe' THEN

rate = 8000.

ELSE rate = 12000.

ELSE

IF room-type = 'Standard' THEN

rate = 3000.

ELSE IF room-type = 'Deluxe' THEN

rate = 5000.

ELSE rate = 8000.

ENDIF

IF stays > 7 :

total-revenue = (rate \* stays) - (rate \* stays

OUTPUT 0.15) + total-revenue. \* 0.15)

OUTPUT. (rate \* stays) - (rate \* stays  
 \* 0.15)

ELSE total-revenue = (rate \* stays) + total-revenue

OUTPUT (rate \* stays)

ENDIF → OUTPUT total-revenue